



SCHOOL *of* DATA SCIENCE

Design a Unique ID Generator

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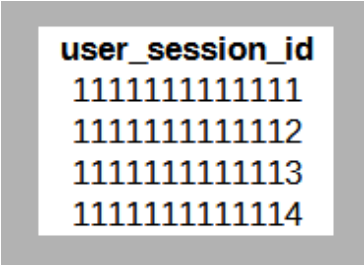
Distributed Computing
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The Problem

Consider: Database table is partitioned and placed on multiple servers (distributed system)

Need to generate and store unique session IDs or user IDs



user_session_id
111111111111
111111111112
111111111113
111111111114

Auto Increment

A database has an auto increment function that can generate sequential numbers.

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Since the data is distributed, this won't work...
there can be duplicates across the servers

server1

user_session_id
111111111111
111111111112
111111111113
111111111114

server2

user_session_id
111111111111
111111111112
111111111113
111111111114

UUIDs

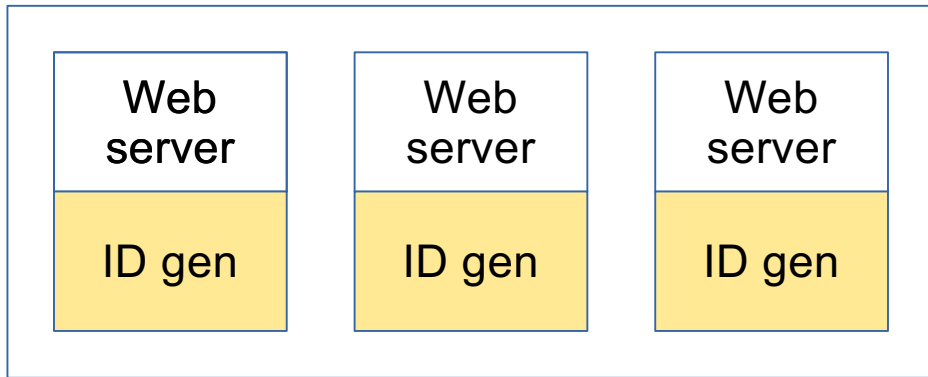
One potential solution – universally unique identifier (UUID)

128-bit number used to identify information in computer systems

Very unlikely to create duplicates

UUIDs can be generated independently without coordination between servers

UUID generator



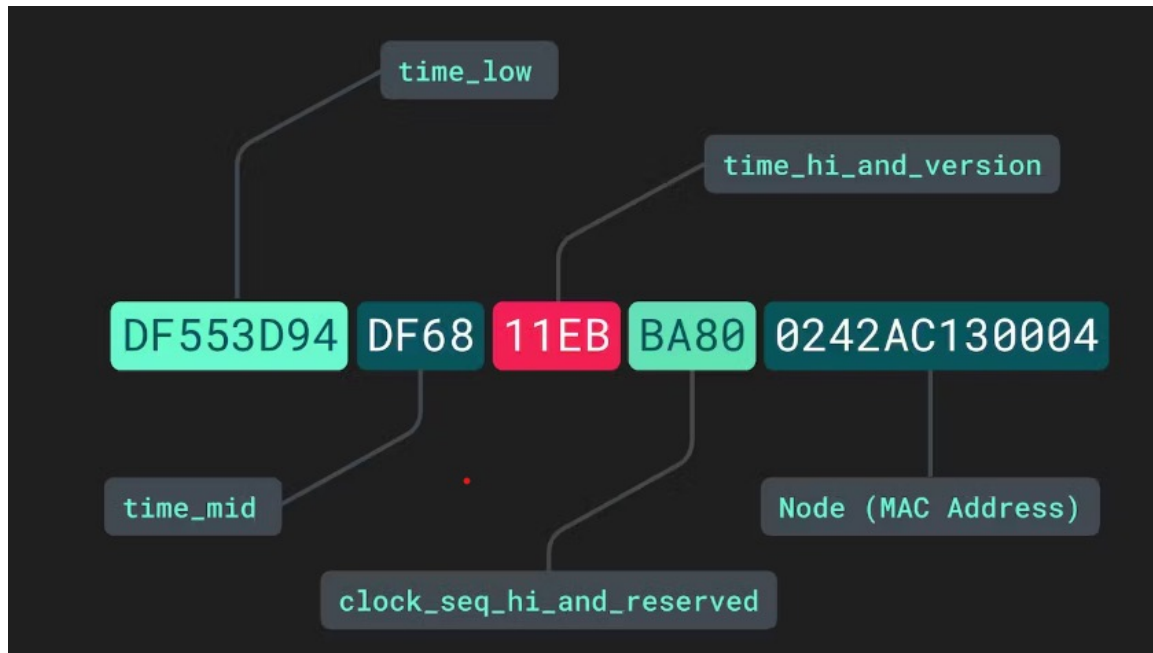
Each server has an ID generator. They operate independently.

UUID examples

```
6ddc8513-dc7b-3b37-b21b-a1ca9440fe14  
8d6a1314-170a-559c-afe7-b68d1d7ee9ac
```

- > Hexadecimal format {0-9, a-f} requires length 32 representation
- > Each character represents 4 bits⁶

Anatomy of UUID



time_low holds first 32 bits of timestamp, for example

UUID Generation – PostgreSQL Example

- > UUID is a subtype of the *string* data type
- > It's being used in the primary key

```
CREATE TABLE users (  
    id UUID NOT NULL DEFAULT gen_random_uuid(),  
    city STRING NOT NULL,  
    name STRING NULL,  
    address STRING NULL,  
    credit_card STRING NULL,  
    CONSTRAINT "primary" PRIMARY KEY (city ASC, id ASC),  
    FAMILY "primary"  
    (id, city, name, address, credit_card)  
);
```

References

System design interview. Alex Xu.

<https://www.mparticle.com/blog/what-is-a-uuid/>